



The Mega2R R package: Tools for accessing and processing common genetic data formats in R.

Daniel E. Weeks^{1,2}, Robert V. Baron¹, Justin R. Stickel¹

¹Department of Human Genetics, Graduate School of Public Health, University of Pittsburgh, ²Department of Biostatistics, Graduate School of Public Health, University of Pittsburgh

Mega2

What is Mega2:

Data reformatting tool in C++ for genetic analysis^{1,2}

Input formats:

PLINK PED/BED	VCF/BCF
Linkage	Impute2
Mega2 Annotated	

Output formats/analysis programs: (Recently added)

Minimac3/Shapeit	VCF
ROADTRIPS	PLINK
Loki	Mendel
PLINK/SEQ	FBAT
Cranefoot	Morgan
Merlin	Beagle
Mega2 Annotated	Simwalk2

Recent enhancements:

- SQLite3 Database
- Alignment to Reference Alleles

SQLite3 Database Tables:

- Pedigree/Person tables
- Locus/Marker/Allele/Map tables
- Phenotype/Genotype tables

Availability:

- Open source (C++) implementation.
- Freely available.
- Linux, Mac, Solaris, and Windows

<https://watson.hgen.pitt.edu/register/>



Mega2->



Mega2R->

References:

1. Baron et al. (2014) Mega2: validated data-reformatting for linkage and association analyses. Source Code Biol Med. 2014 Dec 5;9(1):26.
2. Mukhopadhyay et al. (2005) Mega2: data-handling for facilitating genetic linkage and association analyses. Bioinformatics 21:2556-255.
3. Schaid et al. (2013) Multiple genetic variant association testing by collapsing and kernel methods with pedigree or population structured data. Genet Epidemiology 37(5):408-418.
4. Aulchenko YS et al. (2007) GenABEL: an R library for genome-wide association analysis. Bioinformatics. 2007 May 15;23(10):1294-6.
5. Wu MC et al. (2011) Rare Variant Association Testing for Sequencing Data Using the Sequence Kernel Association Test (SKAT). Am J Hum Genet, 89, 82-93.

Acknowledgments: This work was supported by NIH grant R01 GM076667 (PI: Weeks)

Mega2R

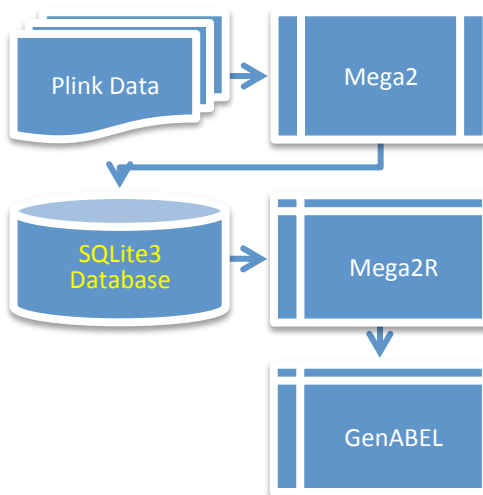
What is Mega2R:

An efficient pathway from many data formats via a **Mega2-created database** into R data frames. These data can be easily used in other R packages and functions.

Input:

Mega2 SQLite3 database tables.

Example conversion to GenABEL:



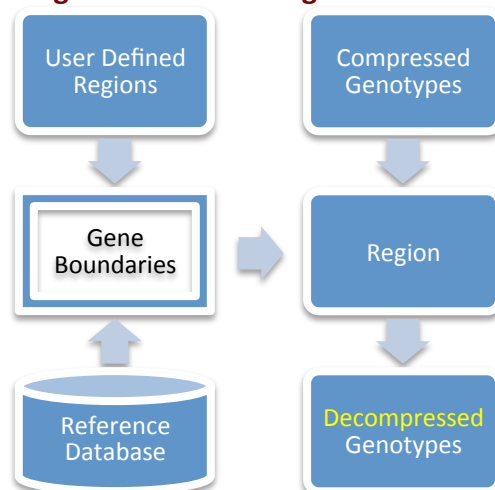
Region extraction:

Memory-efficient extraction of ranges of genotype data within selected regions.

Regions can be defined by the user or by using a transcription database such as those available in Bioconductor.

Genotype data is compressed in the database for storage purposes, Mega2R can extract and decompress data for use in Mega2R's iterative wrapper or for any other R packages.

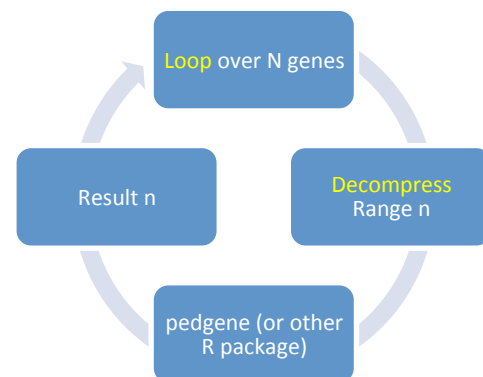
Region extraction diagram:



Iteration wrapper:

Extensive efficient flexible wrapper to iterate through gene regions.

This wrapper can run R functions, which analyze only a single region at a time, on multiple genes.



Example R functions:

Mega2pedgene³
Mega2GenABEL⁴
Mega2VCF
Mega2SKAT⁵

Requirements: R, Mega2

Available with tutorial: CRAN

<https://watson.hgen.pitt.edu/mega2/mega2r/>